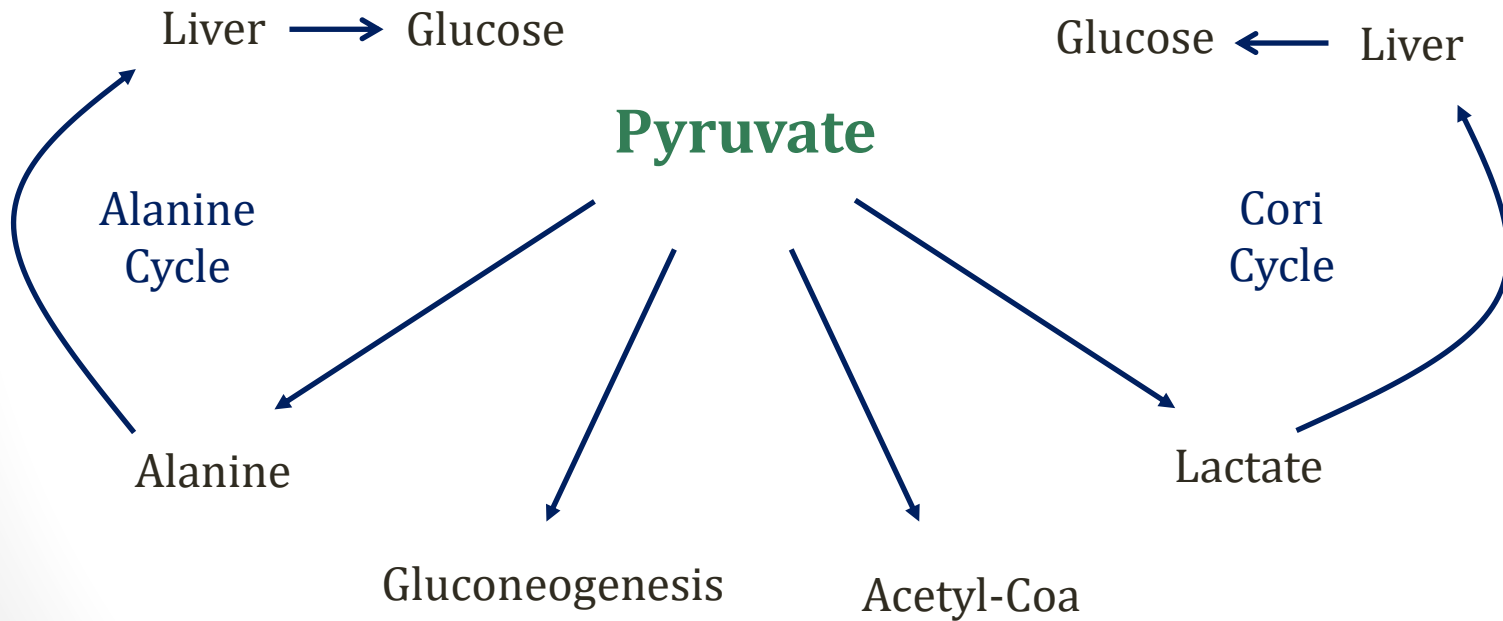


Pyruvate Dehydrogenase

Jason Ryan, MD, MPH

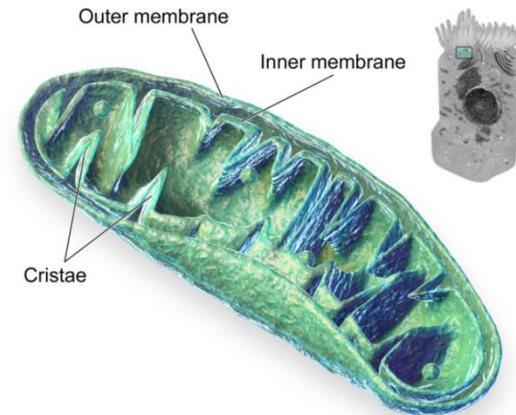
Pyruvate

- End product of glycolysis



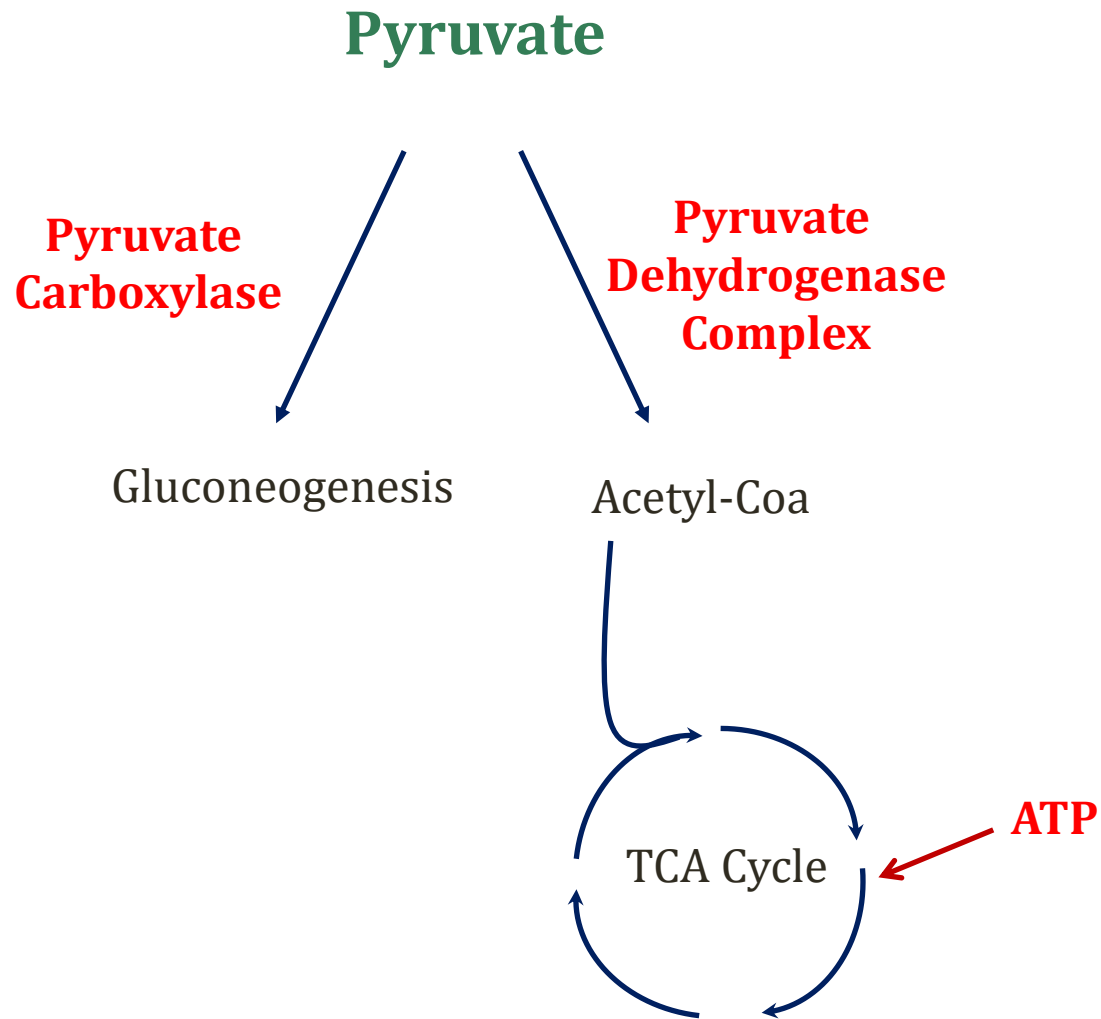
Pyruvate

- Transported into mitochondria for:
 - Entry into TCA cycle
 - Gluconeogenesis
- Outer membrane: a voltage-gated porin complex
- Inner: mitochondrial pyruvate carrier (MPC)



Mitochondria

Pyruvate

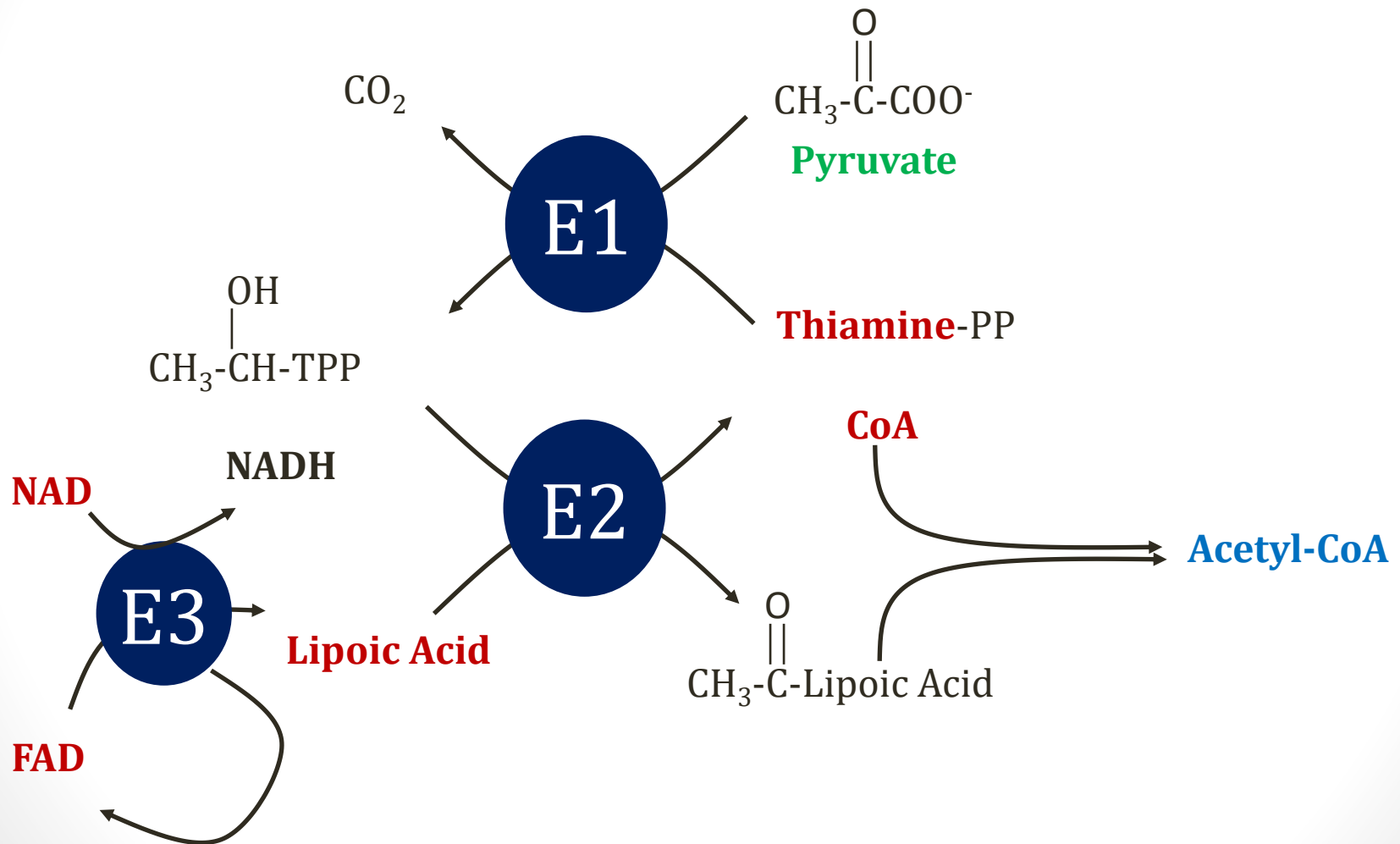


Pyruvate Dehydrogenase Complex

- Complex of 3 enzymes
 - Pyruvate dehydrogenase (E1)
 - Dihydrolipoyl transacetylase (E2)
 - Dihydrolipoyl dehydrogenase (E3)
- Requires 5 co-factors
 - NAD⁺
 - FAD
 - Coenzyme A (CoA)
 - Thiamine
 - Lipoic acid



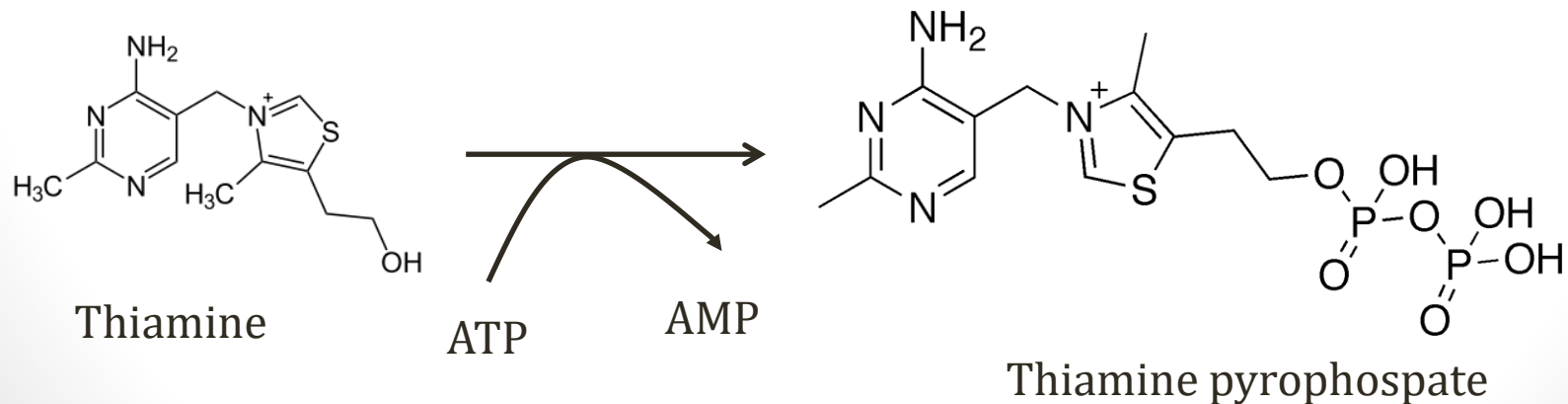
Pyruvate Dehydrogenase Complex



Thiamine

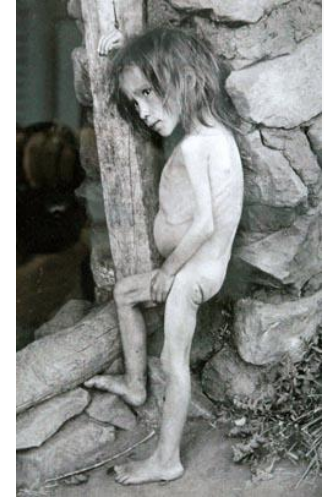
PDH Cofactors

- **Vitamin B1**
- Converted to thiamine pyrophosphate (TPP)
- Co-factor for **four** enzymes
 - Pyruvate dehydrogenase
 - α -ketoglutarate dehydrogenase (TCA cycle)
 - α -ketoacid dehydrogenase (branched chain amino acids)
 - Transketolase (HMP shunt)



Thiamine Deficiency

- ↓ production of ATP
- ↑ aerobic tissues affected most (nerves/heart)
- **Beriberi**
 - Underdeveloped areas
 - Dry type: polyneuritis, muscle weakness
 - Wet type: tachycardia, high-output heart failure, edema
- **Wernicke-Korsakoff syndrome**
 - Alcoholics (malnourished, poor absorption vitamins)
 - Confusion, confabulation



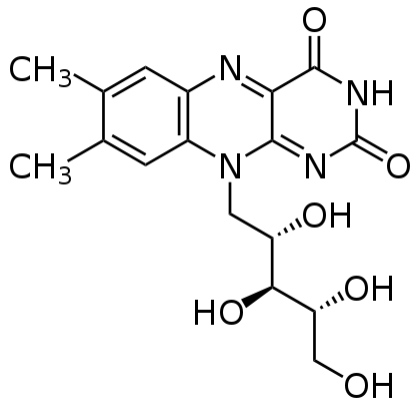
Thiamine and Glucose

- Malnourished patients: ↓glucose ↓thiamine
- If glucose given first → unable to metabolize
- Case reports of worsening Wernicke-Korsakoff

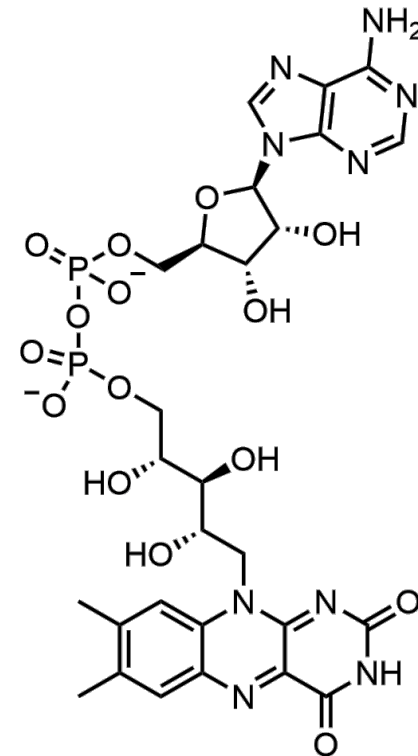
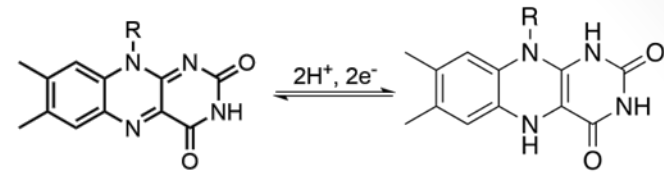
FAD

PDH Cofactors

- Synthesized from **riboflavin (B2)**
- Added to adenosine → FAD
- Accepts 2 electrons → FADH₂



Riboflavin

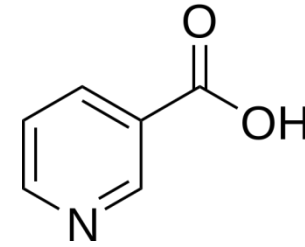


Flavin Adenine
Dinucleotide

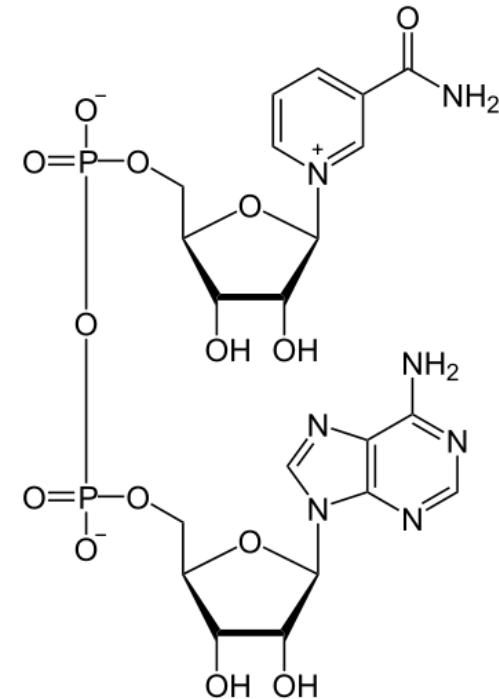
NAD⁺

PDH Cofactors

- Carries electrons as NADH
- Synthesized from **niacin (B3)**
 - Niacin: synthesized from tryptophan
- Used in electron transport



Niacin

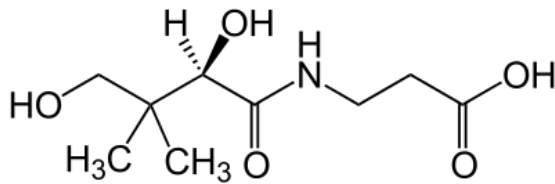


Nicotinamide Adenine
Dinucleotide

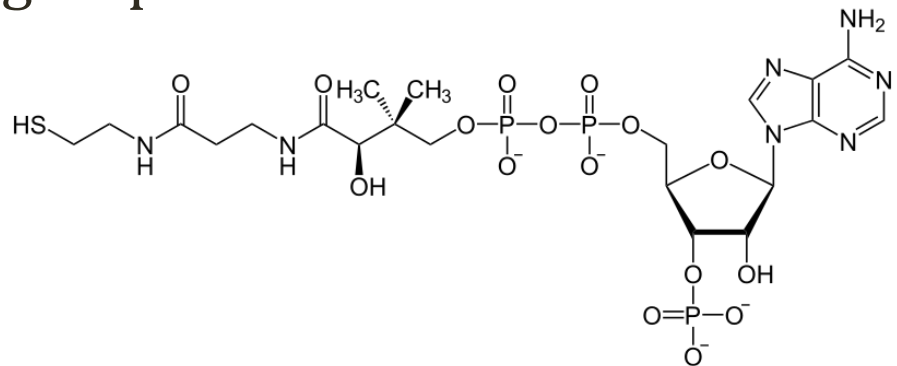
Coenzyme A

PDH Cofactors

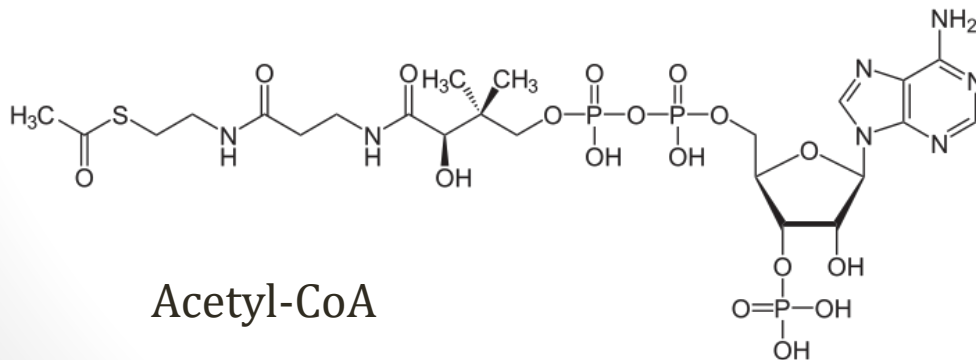
- Also a nucleotide coenzyme (NAD, FAD)
- Synthesized from **pantothenic acid (B5)**
- Accepts/donates acyl groups



Pantothenic Acid



Coenzyme A



Acetyl-CoA

B Vitamins

- B1: Thiamine
- B2: Riboflavin (FAD)
- B3: Niacin (NAD)
- B5: Pantothenic Acid (CoA)

* All water soluble
* All wash out quickly from body
(not stored in liver like B12)



Ragesoss/Wikipedia

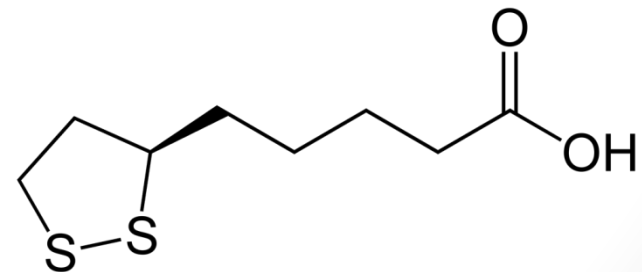
Lipoic Acid

PDH Cofactors

- Bonds with lysine → lipoamide
- Co-factor for E2
- Inhibited by **arsenic**
 - Poison (metal)
 - Binds to lipoic acid → inhibits PDH (like thiamine deficiency)
 - Oxidized to arsenous oxide: smells like **garlic** (breath)
 - Non-specific symptoms: vomiting, diarrhea, coma, death

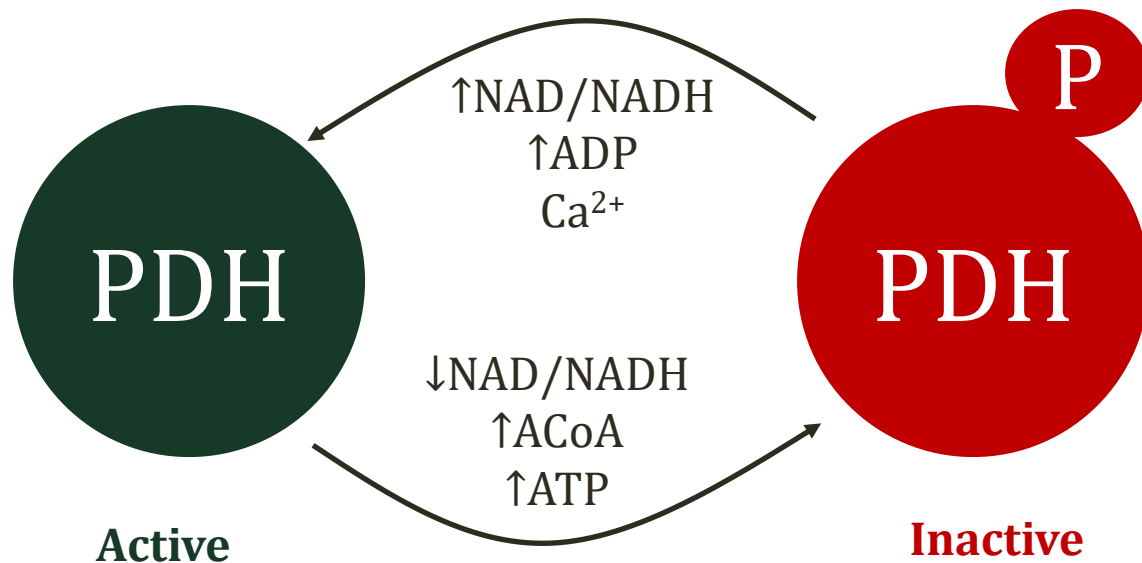


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PDH Regulation

- PDH Kinase: phosphorylates enzyme → inactivation
- PDH phosphatase: dephosphorylation → activation



PDH Complex Deficiency

- Rare inborn error of metabolism
- Pyruvate shunted to alanine, lactate
- Often X linked
- Most common cause: mutations in PDHA1 gene
- Codes for E1-alpha subunit



PDH Complex Deficiency

- Key findings (infancy):
 - Poor feeding
 - Growth failure
 - Developmental delays
- Labs:
 - Elevated alanine
 - Lactic acidosis



Wikipedia/Public Domain

Mitochondrial Disorders

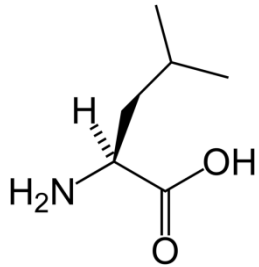
- Inborn error of metabolism
- All cause **severe lactic acidosis**
- Key examples:
 - Pyruvate dehydrogenase complex deficiency
 - Pyruvate carboxylase deficiency
 - Cytochrome oxidase deficiencies

PDH Complex Deficiency

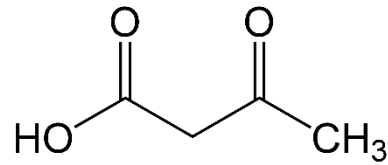
Treatment

- Thiamine, lipoic acid (optimize remaining PDH)
- Ketogenic diet
 - Low carbohydrates (reduces lactic acidosis)
 - High fat
 - Ketogenic amino acids: Lysine and leucine
 - Drives ketone production (instead of glucose)

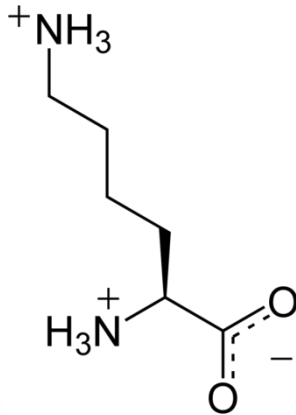
Ketogenic Amino Acids



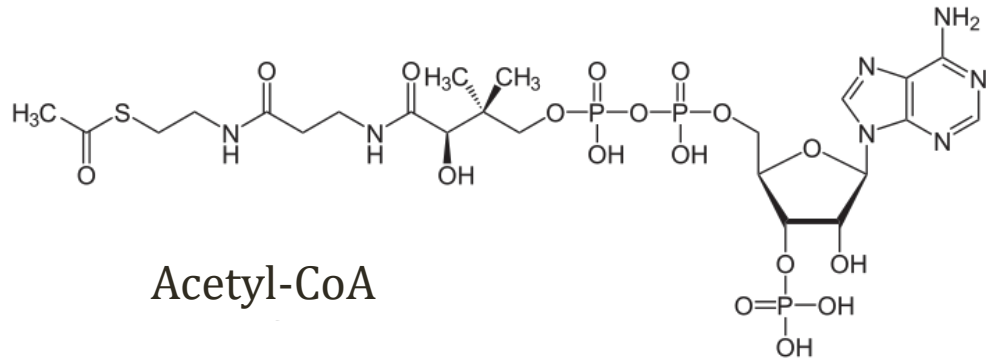
Leucine



Acetoacetate



Lysine



Acetyl-CoA